

Stellarocq: Machine-Checked Certificates for Stellarator Equilibria

Charles C. Norton
cnorton2@binghamton.edu

Abstract

Stellarator design rests on ideal-MHD equilibria computed by VMEC and its descendants, whose output is a file of floating-point Fourier coefficients. Stellarocq is a Rocq development that turns questions about such a file into theorems about the magnetic field it defines. The field is reconstructed from the exact numbers in a VMEC++ output file by VMEC’s own half-grid rule, force balance, harmonics, flux-surface integrals, the Mercier criterion, quasisymmetry and free-boundary balance are expressions over that reconstruction, and each claim is decided by a checker extracted from a proof that its verdict implies the claim. On VMEC++ output the certificates attribute a converged residual to mode truncation, expose an orientation test that is not invariant under the poloidal rotation VMEC applies to asymmetric boundaries, measure the error of VMEC’s radial scheme on manufactured problems to twelve digits, certify a resonant harmonic that radial refinement does not remove, and bound the free-boundary pressure jump for every vacuum pressure between the answers of NESTOR and BIEST.

1 Introduction

The Variational Moments Equilibrium Code (VMC) [1, 2] computes three-dimensional ideal-MHD equilibria with nested flux surfaces by minimizing the plasma energy over Fourier representations of the surfaces. It and its reimplementations VMEC++ [3, 4] sit at the centre of stellarator design and optimization. A run ends with a *wout* file: Fourier coefficients of the surfaces on a radial grid, profiles, and derived quantities such as the Mercier stability terms, all in binary64 floating point.

What such a file establishes about the field it describes is usually a matter of convergence metrics internal to the solver and of diagnostics recomputed in floating point. Stellarocq states it as a theorem. The object of every statement is the field reconstructed from the exact numbers in the file by the rule the solver itself uses. A claim about that field (a bound on the force residual over a region, the enclosure of a harmonic, the sign of a Mercier term, the existence of a discrete equilibrium in a box) is decided by a checking function written in Rocq [5], proven to imply the claim when it returns `true`, and extracted to OCaml [6] to run at the scale a three-dimensional equilibrium needs. Real-number semantics and interval arithmetic come from Coquelicot [7], Flocq [8] and CoqInterval [9].

Section 2 states the certified object, Section 3 the certificate forms and their soundness theorems, and Section 4 what they establish on VMEC++ output. The part of the development described here is 29 files and about 22,100 lines of Rocq; it is available at <https://github.com/CharlesCNorton/stellarocq>.

2 The certified object

Reconstruction. VMEC keeps the surface coefficients of R and Z on the full radial grid $s_j = jh$ and the stream function λ , the rotational transform ι and every field quantity on the half grid

$s_{j+1/2}$. Stellarocq's `Physics.v` writes the rule that connects them. On a half point s_h between nodes a and b , a coefficient of even poloidal mode number is the mean of its two node values and one of odd mode number is averaged with the $\sqrt{s}$ weight VMEC folds into its Jacobian,

$$c_k(s_h) = \frac{1}{2}(c_k(a) + c_k(b)) \quad (m_k \text{ even}), \quad c_k(s_h) = \frac{1}{2}\sqrt{s_h} \left(\frac{c_k(a)}{\sqrt{s_a}} + \frac{c_k(b)}{\sqrt{s_b}} \right) \quad (m_k \text{ odd}), \quad (1)$$

with radial derivatives the matching difference quotients (for odd m_k plus the derivative of the $\sqrt{s}$ factor), and the angular derivatives exact derivatives of the series $R = \sum_k c_k \cos(m_k u - n_k v)$, $Z = \sum_k z_k \sin(m_k u - n_k v)$. The half-grid λ and ι are the file's own, Φ' is constant, and the pressure is the input's profile. For statements at an arbitrary radius the half-point values and slopes are joined by a cubic Hermite interpolant, which agrees with the half-grid rule at the half points.

Field and residual. With $\sqrt{g} = R(R_u Z_s - R_s Z_u)$, the field is $B^u = \Phi'(\iota - \lambda_v)/\sqrt{g}$, $B^v = \Phi'(1 + \lambda_u)/\sqrt{g}$, its covariant components $B_i = g_{iu} B^u + g_{iv} B^v$, and $\mu_0 \sqrt{g} J^s = \partial_u B_v - \partial_v B_u$. The residual of $\mathbf{J} \times \mathbf{B} = \nabla p$ has the components

$$r_u = -\mu_0 \sqrt{g} J^s B^v, \quad r_v = \mu_0 \sqrt{g} J^s B^u \quad (\text{at the outer half point}), \quad (2)$$

$$r_s = (\partial_v B_s - \partial_s B_v) B^v - (\partial_s B_u - \partial_u B_s) B^u - \mu_0 dp/ds \quad (\text{at the node}), \quad (3)$$

where at the node radial derivatives are centred differences of half-grid values and values are their averages, as in VMEC. These are `expr` trees, which `Physics.v` states without proofs, and the development proves what they compute. Read at a free radius, with every radial derivative that of the Hermite reconstruction, they form the continuum residual, whose three outputs are the covariant components along $\mathbf{e}_s$, $\mathbf{e}_u$ and $\mathbf{e}_v$ of $\mu_0(\mathbf{J} \times \mathbf{B} - \nabla p)$ of the reconstructed field, with $\mu_0 \mathbf{J}$ the curl of that field computed in Cartesian coordinates (`continuum_force`, and `continuum_force_asym` for a state without stellarator symmetry, from `Force.force_law`). Read at a node, the angular outputs are those formulas at the outer half point and the radial output is VMEC's combination of the fields at the two half points, each field that of the series with VMEC's half-grid values and slopes (`node_residual`, `node_residual_asym`).

Identities of the reconstruction. Properties a floating-point code can only sample are theorems here. The reconstructed field is exactly solenoidal (`divergence_free`); the $(0,0)$ coefficient of λ drives nothing (`lambda_gauge`), which exhibits the poloidal relabelling as an exact kernel direction of the linearized problem; the residual has no component along the field (`residual_along_field`); and the Euler–Lagrange expressions of VMEC's energy $W = \int (B^2/2 - \mu_0 p) \sqrt{g}$ at fixed λ , ι , flux and pressure are $-\sqrt{g}$ times the cylindrical components of the force (`energy_gradient_is_force`), so that the forces the solver drives to zero are the ideal-MHD force. `Symmetry.v` proves that a quasisymmetric field makes the two-term ratio $(\mathbf{B} \times \nabla s \cdot \nabla B)/(\mathbf{B} \cdot \nabla B)$ a flux function and the triple product $\nabla s \cdot (\nabla B \times \nabla(\mathbf{B} \cdot \nabla B))$ vanish, which is the direction a certificate refuting quasisymmetry rests on. Physical assumptions, such as second-order consistency of the discretization, are propositions in `Hypotheses.v` carried as explicit hypotheses of the theorems that use them, never as axioms.

3 Certificates and their soundness

A certificate is a text file that a Python generator in `gen/` writes from a wout file. It carries the file's numbers and the bounds it claims, and the extracted checker recomputes every enclosure from those numbers, so nothing the generator computes is trusted.

Expressions and intervals. `Expr.v` gives `expr` two meanings: a value over extended reals, in which division by zero and the square root of a negative number give `Xnan`, and a value over CoqInterval’s floating-point intervals. `ieval_correct` states that the interval contains the extended-real value. A bounded output interval therefore proves both that the value is a real number and that it lies in the bounds. Subexpressions shared between the residual components are bound once to environment slots, and `iextend_correct` states that extending both environments by the same bindings preserves containment.

Points. A point certificate carries, for every environment slot, the integer mantissa and power-of-two exponent of a double of the wout file, so no rounding happens between the file and the certificate, together with claimed bounds $N2^q$ on the three residual components. Its soundness theorem reads:

```

Definition point_sound (c : cert) (p : cpoint) : Prop :=
  let r3 := residual (pt_es p) (c_cfg c) (c_modes c) in
  let env := point_env c p in
  (exists x, xeval env (r_s r3) = Xreal x /\
    Rabs x <= IZR (c_NS c) * powerRZ 2 (c_qS c)) /\ ...
Theorem check_cert_correct : forall c,
  check_cert c = true -> Forall (point_sound c) (c_points c).

```

Cells and coverings. `Deriv.v` differentiates expressions symbolically and proves the chain rule for the result in CoqInterval’s extended-real sense. A cell certificate bounds a component over a rectangle of two varied slots (two angles, or an angle and the radius) by the mean-value combination of its centre value and interval enclosures of its derivatives over the rectangle (`check_ccert_correct`). Reversed, the same data proves a component bounded away from zero, so that no field of the reconstructed form balances in the cell. That the cells of a certificate leave no gap is not assumed from the generator that wrote them: `Cover.v` checks it as a chain of integer inequalities, so a certificate claiming a range it does not cover is rejected like one whose bounds are too small.

Harmonics and integrals. A spectral solution balances forces mode by mode over the modes it retains, so the quantity it makes small is the discrete Fourier harmonic of the residual at a retained mode, a finite sum $\sum_i r(p_i) \cos(mu_i - nv_i)$ over a uniform grid of angles that `Project.v` encloses (`harm_encloses`). `TrigPoly.v` and `Harmonic.v` integrate exactly the components whose trigonometric degree lies below the point counts of an equispaced grid, and `Quad.v` encloses other angular integrals cell by cell. `Mercier.v` assembles DShear, DCurr, DWell, DGeod and DMerc as VMEC’s `mercier.f90` defines them [10, 11] from enclosed integrals, inside the extracted code, so that a verdict is a theorem about the criterion rather than about its ingredients. DGeod is the Cauchy–Schwarz defect of three surface integrals and is proven never positive, whatever their enclosures (`dgeod_nonpositive`).

Boxes of states. In `Box.v` every input slot carries a half-width, so one interval evaluation covers every field whose coefficients lie in a box. A floor certified over a box cannot be an artefact of the one state at which a solver stopped.

Discrete equilibria. `Newton.v` proves that a passing interval Newton test in Krawczyk form [12, 13] on any system of expressions gives exactly one zero in its box: the map $x - AF(x)$ contracts the box into itself, an approximate inverse of A makes its fixed point a zero of F , and the contraction

Case	Certified quantity	Result
<code>li383_low_res</code>	residual over cells	does not fall from 16 to 121 surfaces; falls more than twentyfold from 25 to 98 modes
<code>up_down_asym</code> , $s = 0.125$, 16 surfaces	Mercier current term	-3.7×10^{-3} against the file's -6.1×10^{-4} , reversing DMerc; both tend to zero from 33 to 257 surfaces with the file's sign
Solov'ev, 1–3 surfaces	collocated residual	exactly one zero in boxes of radius 5.7×10^{-10} to 6.3×10^{-9}
manufactured, 3D	$\max x_h - x^* $, 9/17/33/65 surfaces	$2.73432936716 \times 10^{-6}$, $7.16912169671 \times 10^{-7}$, $1.83032584737 \times 10^{-7}$, $4.60748183004 \times 10^{-8}$
manufactured, asymmetric rippled tokamak, $\iota = 1/2$ at $s = 0.625$	$\max x_h - x^* $, 9/17/33 surfaces (2, 1) harmonic of r_s , 65/129/257 surfaces	$2.69073057006 \times 10^{-5}$, $7.20667052904 \times 10^{-6}$, $1.86266877275 \times 10^{-6}$ [8.35, 8.45], [8.67, 9.07], [6.19, 7.76] $\times 10^{-5}$; the (2, 0) and (1, 0) harmonics fall 3.8 to 5.3-fold per doubling
CTH-like, free boundary	jump of $p + B^2/2$ against $B_{\text{vac}}^2/2$	floor at NESTOR's largest jump, ceiling over NESTOR's grid for every vacuum pressure between NESTOR's and BIEST's: [7.556, 7.560] $\times 10^{-4}$ at 25 surfaces, [4.9648, 4.9672] $\times 10^{-4}$ at 385, each ceiling below the floor before it
<code>nfp4_QH_ns50</code>	two-term quasisymmetry ratios	span 1.7 at mid radius (Solov'ev: 2×10^{-4}), unchanged from 50 to 162 modes

Table 1: Certified results on VMEC++ equilibria.

makes that zero unique. `Colloc.v` collocates the residual of `Physics.v` over a band of surfaces as such a system, with the surface coefficients as unknowns (`colloc_correct`). `Wide.v` evaluates the system's outputs at the centre of the box in floating point of any precision.

Trust base. The kernel checks the proofs; what runs is the OCaml extraction of the checking functions, linked with an unverified driver that parses certificates into the extracted data types and prints verdicts. The trusted components are therefore the Rocq kernel and extraction, the OCaml toolchain, the implementation of Rocq's primitive 63-bit integers and binary64 floats, the driver's parser, and the correspondence between a certificate's dyadic numbers and the doubles of its wout file, which a separate tool, `gen/verify_cert.py`, compares. For the soundness theorems above, `Print Assumptions` reports the two axioms of the standard library's classical Dedekind reals, excluded middle, functional extensionality and the specifications of the primitive operations. The build pins Rocq 9.1.1, CoqInterval 4.11.4, Flocq 4.2.2, Coquelicot 3.4.5 and OCaml 4.14.2.

4 Results on VMEC++ output

Table 1 collects the results; this section states what each establishes. All equilibria are VMEC++ runs or files from its test data, and every number reported as certified is an enclosure a passing checker verdict proves.

Residuals and stability terms at low resolution. The certified residual of `li383_low_res` is insensitive to radial refinement and falls with the mode set (Table 1): what a converged VMEC++ run leaves at this resolution is the truncation of the mode set, which the solver's convergence

measure, taken over the retained modes, does not see. At $s = 0.125$ of `up_down_asym` with sixteen surfaces, the certified current term of the Mercier criterion is -3.7×10^{-3} where the file's, formed from difference quotients across surfaces, is -6.1×10^{-4} , and the difference reverses the sign of DMerc; re-solved at 33 to 257 surfaces the two fall toward zero together and the file's sign survives. At coarse resolution the verdict on local stability depends on how the radial derivatives are taken.

An orientation test that is not gauge invariant. Four invariance tests of VMEC++ have expected answers that the development fixes: an axisymmetric input solved with and without toroidal modes (every toroidal derivative of an axisymmetric reconstruction is the zero expression), a stellarator-symmetric input solved through the symmetric and the asymmetric code paths (a series whose coefficients vanish contributes nothing), a restart with a shifted $(0,0)$ coefficient of λ (`lambda_gauge`), and a five-period stellarator solved over one period and over the torus (the reconstruction reads each mode by its absolute toroidal number). The second exposed an orientation test that is not invariant under the poloidal rotation VMEC applies to an asymmetric boundary: the test multiplies two $m = 1$ coefficients whose product the rotation can drive to zero, where the invariant is the $m = 1$ determinant $r^{cc}z^{sc} - r^{sc}z^{cc}$. Relabelling the `up_down_asym_current` boundary by $\theta \rightarrow \pi - \theta$ sends educational_VMEC to its iteration cap at $f_{sqr} = 2.07 \times 10^{-2}$ instead of converging to 5.07×10^{-12} . It was reported as vmecpp#788 [14], after which VMEC++ tests the determinant [15], and as educational_VMEC#27 [16].

Discrete equilibria and the order of the scheme. One, two and three surfaces of the Solov'ev equilibrium [22] each have exactly one zero of the collocated residual in a box shaped to the coefficients, of radius 5.7×10^{-10} to 6.3×10^{-9} , around a centre that Newton's method moved 2.2×10^{-6} to 3.5×10^{-6} from the file's coefficients; with five surfaces the test does not pass. `HalfGrid.v` proves the node rule second-order consistent with the residual of the continuously reconstructed field (`node_consistent`) and derives the second-order bound that `discretization_is_consistent` states from consistency and a stability hypothesis (`lax_second_order`); near the axis, where the innermost interval is read linearly, the rule is proven first order in the radial derivatives. Manufactured solutions [23] measure the order directly. A manufactured mapping on the annulus $1/4 < s < 1$ is not an equilibrium; its continuum residual at the collocation points is a source f , and the zero x_h of $r - f$ over the interior nodes, with λ given so that the problem sits on the quotient by the poloidal gauge, is the discrete solution. The interval Newton test puts exactly one zero in a box 5×10^{-18} wide per unknown around a centre refined on outputs read at 128 bits, so the error $\max |x_h - x^*|$ is known to twelve digits (Table 1). It falls by 3.81, 3.92 and 3.97 per doubling for a three-dimensional mapping and by 3.73 and 3.87 for one without stellarator symmetry (Figure 1a), and it is unchanged at five times the pressure, which enters the radial force only as a flux function the rule and its source share.

A forced current sheet. On VMEC++'s circular tokamak ($R_0 = 6$, $a = 2$) with $\iota = 0.9 - 0.64s$, which is $1/2$ at $s = 0.625$, and a $(1,1)$ boundary ripple of amplitude 0.03, toroidal coupling drives a $(2,1)$ harmonic of the radial force on the rational surface. Its certified value, the rectangle rule for the integral of $r_s \cos(2u - v)$ on a 64×32 grid of angles, enclosed by `dharm_correct`, stays between 6.2×10^{-5} and 9.1×10^{-5} from 65 to 257 surfaces and is linear in the ripple; the enclosure at 65 surfaces holds for every state within a relative 10^{-12} of the converged coefficients. On the same surface the axisymmetric harmonics $(2,0)$ and $(1,0)$ fall by 3.8 to 5.3 per doubling, the order of the scheme, while the $(3,1)$ sideband falls by only 1.2 to 1.3 (Figure 1b), and the same ripple on a transform profile with no low-order rational surface nearby leaves nothing behind. This is

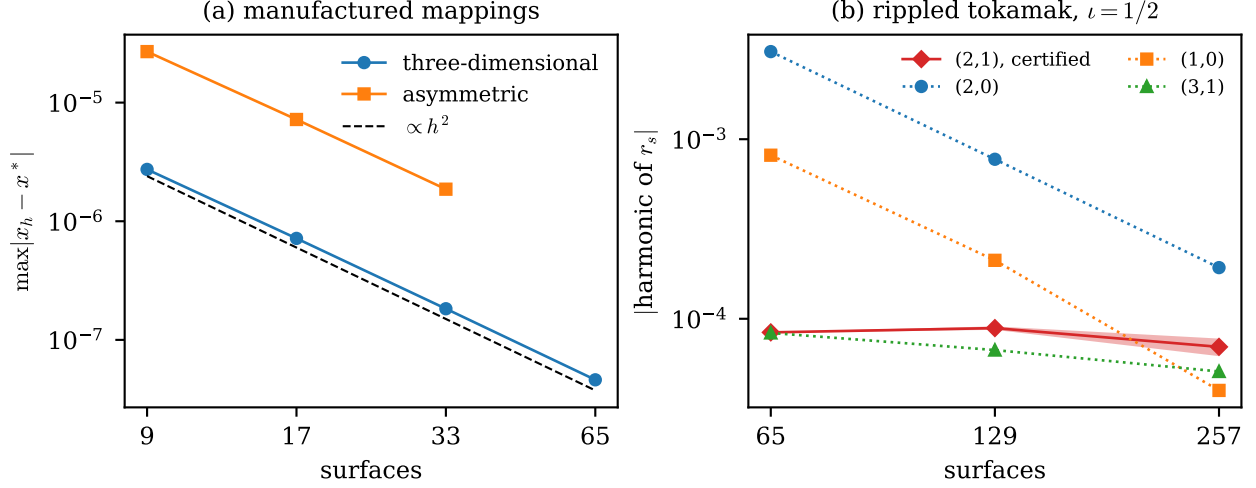


Figure 1: (a) The certified error of the half-grid scheme on the manufactured mappings; the dashed line falls as h^2 . (b) Harmonics of the radial force r_s on the surface $\iota = 1/2$ of the rippled tokamak: the certified enclosures of the resonant $(2, 1)$ harmonic, shaded, and the floating-point values of the $(2, 0)$, $(1, 0)$ and $(3, 1)$ harmonics of the same reconstruction. The first two fall 3.8 to 5.3-fold per doubling; the resonant harmonic and its $(3, 1)$ sideband stay.

the obstruction a nested-surface field meets at a rational surface [24, 25], whose ideal response is a current sheet. In SPEC [26], two Beltrami volumes separated by an ideal interface at the rational surface carry the $(2, 1)$ sheet as a jump of 1.47×10^{-5} in the covariant poloidal field at ripple 0.03, as in the equilibria of [27], and with the rational surface inside a relaxed volume SPEC’s field-line tracing finds a $(2, 1)$ island whose width grows from 0.107 to 0.170 in the volume’s radial coordinate as the ripple goes from 0.0075 to 0.03.

Free-boundary balance. At a free-boundary equilibrium $p + B^2/2$ is continuous across the plasma boundary, where outside it is the pressure of the full vacuum field. VMEC++ computes that field with NESTOR [28, 2]; BIEST’s virtual-casing field [29, 30] gives it independently. The certificate sweeps the vacuum pressure over the segment between the two answers, $P_{\text{vac}} = P_{\text{mid}} + t \delta P$ with $t \in [-1, 1]$, and the plasma side is VMEC’s extrapolation of the two outermost half points to $s = 1$. On the CTH-like test case, NESTOR’s jump where it is largest is certified at least 7.556×10^{-4} at 25 surfaces, and the jump at every point of NESTOR’s grid at most 7.560×10^{-4} for every t . At 49, 97, 193 and 385 surfaces each such ceiling lies below the floor of the resolution before it, down to $[4.9648, 4.9672] \times 10^{-4}$ at 385, so the certified jump falls under radial refinement. Covered over the whole boundary by 512×128 cells per field period, with the vacuum side the trigonometric interpolant of each answer’s grid, it is at most 8.5×10^{-3} at 25 surfaces and 1.05×10^{-2} at 385: between grid points the bound is set by that interpolant, whose angular resolution radial refinement leaves as it is.

Quasisymmetry. A field is quasisymmetric exactly when the two-term ratio is a flux function [31, 32]. For a quasisymmetric field every discrete harmonic of the defect of the claim “the ratio equals F_0 ” is one number times the matching harmonic of its second term, so two harmonic ratios with disjoint enclosures refute it (`two_term_refuted`). On `wout_nfp4_QH_ns50` the certified ratios span 1.7 at mid radius, where the axisymmetric Solov’ev equilibrium, quasisymmetric by construction,

spans 2×10^{-4} from the reconstruction’s own residual; the span is unchanged from 50 to 162 modes. The field of that file is not quasisymmetric on any surface tested.

Use in VMEC++. The certified values enter a branch of VMEC++ [17] as regression tests whose tolerances are the enclosures: the manufactured-solution errors, the resonant floor with the fall of the axisymmetric harmonics, and the free-boundary jump, which reads NESTOR’s boundary vacuum field through an option proposed upstream [18]. Tests that follow from theorems here are part of VMEC++: a finite difference of its energy equals its unpreconditioned forces contracted with the perturbation, the discrete form of `energy_gradient_is_force` [19], and three of the invariances above hold [20]. Tests that DGeod is never positive and that the energy Hessian is symmetric are proposed [21].

5 Related work

Machine-checked numerical analysis has covered the discretization and the floating-point program of the one-dimensional wave equation [33], verified ODE enclosures up to the Lorenz attractor [34], and the checking of large certificates in the formal proof of the Kepler conjecture [35]. Computer-assisted proofs for partial differential equations are well developed outside proof assistants [36]. In plasma physics, equilibrium codes are verified by convergence studies, manufactured solutions and comparison between codes such as VMEC and SPEC [26, 27]. We know of no prior work that states the output of a stellarator equilibrium code as a machine-checked theorem about the field it defines.

6 Conclusion

Stellarocq gives the output of an equilibrium code a meaning a proof assistant can check: bounds, harmonics, stability terms, quasisymmetry verdicts and the existence of discrete solutions, each a theorem about the field reconstructed from the file’s exact numbers. The same machinery extends from the equilibrium to its coil field, where it carries a machine-checked KAM argument for the flux surfaces of a real stellarator, the subject of separate work.

Availability. The development, the certificate generators and the scripts that draw Figure 1 are at <https://github.com/CharlesCNorton/stellarocq>, commit 517aaff; `make all` builds the proofs and the checker, `make audit` prints the assumptions of every theorem, and the release `checker-74169ad` carries the same checker linked statically for x86_64 Linux. The VMEC++ tests that carry the certified values are on the branch of [17].

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